Industry 4.0 Predictive Maintenance System

PyTorch 1D-CNN + Siamese Network Telemetry Platform · 18 Hydraulic Channels

System Architecture · 1D-CNN Mechanics · 18-Sensor Catalog · Roadmap · Algorithms & Models Inventory

1. Executive Summary & Project Purpose

Project Overview: In automated manufacturing, high-pressure hydraulic sub-assemblies (pumps, valves, actuators, and cylinders) operate under heavy load and continuous duty cycles. Unexpected breakdowns lead to catastrophic equipment failures, costly factory downtime, and operator hazards. This project implements a production-grade **Predictive Maintenance (PdM) platform** that continuously monitors multi-channel sensor telemetry, models temporal signatures, detects degradation patterns early, and flags equipment anomalies *before* functional breakdown occurs.

Dual Deep Learning Paradigm: Industrial telemetry features both known repetitive faults and unpredictable anomalies. The project pairs two complementary PyTorch networks: (1) a **1D Convolutional Neural Network (1D-CNN)** that performs high-speed binary fault classification (Normal: 0 vs. Fault: 1), and (2) a **Siamese Metric-Learning Network** that projects operational cycles into a 128-dimensional embedding space, calculating pairwise Euclidean distance against healthy baseline cycles for few-shot anomaly detection.

2. Deep Dive: How the 1D-CNN is Used in the Project

The 1D-CNN (implemented in `src/cnn_model.py`) serves as the primary real-time fault detector. Unlike standard 2D-CNNs designed for spatial image grids, **1D-CNNs slide convolutional filters across time steps**, capturing temporal dynamics, rate-of-change, and inter-sensor correlations across all 18 hydraulic telemetry channels simultaneously.

Input Representation & Temporal Windowing:

• **Tensor Shape:** The network operates on temporal sliding windows of shape (Batch Size, Sequence Length = 8, Features = 18).
• **Channel Transposition:** In forward(x), the tensor is permuted to (Batch Size, Channels = 18, Time = 8). The 18 hydraulic sensors act as input channels and 8 consecutive time ticks form the temporal receptive field.
• **Single-Tick Tiling for Inference:** When individual sensor readings arrive via the FastAPI or Streamlit UI, the reading is tiled across 8 timesteps to form a stationary synthetic window (1, 8, 18), guaranteeing identical batch and streaming inference pipelines.

Detailed Neural Architecture:

1. Three Hierarchical 1D-Convolutional Feature Blocks:

- *Block 1:* Conv1d(in=18, out=32, kernel=3, padding=1) → BatchNorm1d(32) → ReLU() → MaxPool1d(2, stride=2) [Time: 8 → 4]
- *Block 2:* Conv1d(in=32, out=64, kernel=3, padding=1) → BatchNorm1d(64) → ReLU() → MaxPool1d(2, stride=2) [Time: 4 → 2]
- *Block 3:* Conv1d(in=64, out=128, kernel=3, padding=1) → BatchNorm1d(128) → ReLU() → MaxPool1d(2, stride=2) [Time: 2 → 1]

Engineering Role: Kernels automatically learn filter weights that detect physical phenomena: pressure surges, current spikes, phase delays between displacement and power, and solenoid voltage drops. BatchNorm stabilizes activation distributions, and MaxPool1d reduces temporal noise.

2. Fully Connected Dense Classification Head:

- *Flattening:* Collapses feature maps into a 128-dimensional representation (128 channels × 1 timestep).
- *Dense Layers:* Linear(128, 256) → ReLU() → Dropout(0.5) → Linear(256, 128) → ReLU() → Dropout(0.5) → Linear(128, 2).

Engineering Role: 50% Dropout mitigates co-adaptation and prevents overfitting on specific operating cycles. Output layer generates raw 2-class logits.

3. Loss Optimization & Inference Confidence:

The model is trained via nn.CrossEntropyLoss using the Adam optimizer (learning rate = 0.001) with early stopping. In production inference, Softmax transforms output logits into normalized class probabilities: Class 0 denotes **Normal**, and Class 1 denotes **Fault**. The highest probability is surfaced as the confidence metric.

3. Sensor Features Catalog: 18 Telemetry Channels Explained

The industrial hydraulic system monitors 6 sub-assemblies (BLO, BHL, BHR, BRU, HR, HL). Each module captures 3 signals: **Weg** (actuator displacement current in mA), **Power** (active wattage in W), and **Voltage** (excitation potential in V). Below is the complete feature dictionary detailing why each sensor channel is monitored:

Feature Column Name	Sub-Assembly & Signal Type	Why It Is Used (Engineering Explanation)
I_w_BLO_Weg	Bottom Left Outlet · Displacement	Measures actuator stroke current to detect mechanical travel delays and sticking in the bottom-left outlet valve.
O_w_BLO_power	Bottom Left Outlet · Power	Monitors electrical power draw to detect excess motor workload, mechanical resistance, or valve binding.
O_w_BLO_voltage	Bottom Left Outlet · Voltage	Tracks operating supply voltage to identify line voltage drops, power instability, or solenoid coil degradation.
I_w_BHL_Weg	Bottom Left Hydraulic · Displacement	Measures cylinder piston position current to detect stick-slip motion or uneven mechanical stroke travel.
O_w_BHL_power	Bottom Left Hydraulic · Power	Measures active power consumption to identify pressure drops, internal fluid bypass, or hydraulic pump wear.
O_w_BHL_voltage	Bottom Left Hydraulic · Voltage	Monitors actuator control voltage to catch solenoid switching lag and control circuit voltage degradation.
I_w_BHR_Weg	Bottom Right Hydraulic · Displacement	Tracks linear stroke displacement current to verify synchronized cylinder movement with the left hydraulic bank.
O_w_BHR_power	Bottom Right Hydraulic · Power	Monitors active power draw to spot excessive mechanical resistance, hydraulic fluid throttling, or blockages.
O_w_BHR_voltage	Bottom Right Hydraulic · Voltage	Captures terminal voltage levels to identify loose wiring connections, contact resistance, or supply dips.
I_w_BRU_Weg	Bottom Right Upper · Displacement	Tracks upper actuator displacement current to verify proper valve seat engagement and stroke travel limits.
O_w_BRU_power	Bottom Right Upper · Power	Quantifies power draw to detect motor overload, fluid contamination drag, or hydraulic cylinder seal friction.
O_w_BRU_voltage	Bottom Right Upper · Voltage	Records drive voltage across the upper actuator to verify reliable solenoid energization across cycles.
I_w_HR_Weg	Hydraulic Right · Displacement	Measures main right cylinder displacement current to verify structural alignment and full hydraulic stroke cycle.
O_w_HR_power	Hydraulic Right · Power	Tracks main right actuator power consumption to identify relief valve leaks, pump slip, or pressure loss.
O_w_HR_voltage	Hydraulic Right · Voltage	Monitors main right valve operating voltage to detect power fluctuations and solenoid thermal degradation.
I_w_HL_Weg	Hydraulic Left · Displacement	Measures main left cylinder displacement current to detect mechanical misalignment and cylinder seal wear.
O_w_HL_power	Hydraulic Left · Power	Measures left actuator wattage to detect hydraulic cavitation, mechanical binding, or fluid viscosity changes.
O_w_HL_voltage	Hydraulic Left · Voltage	Tracks left hydraulic valve supply voltage to identify electrical brownouts and control line degradation.

Dataset Meta-Attributes: Timestamp (cyclical index partitioning continuous high-frequency measurements into operational duty cycles) and Labels (ground truth binary target: 0 = Normal / Healthy, 1 = Fault / Anomaly).

4. Industrial Value & Target Deployment Sectors

Predictive maintenance delivers immediate operational ROI by transforming plant strategy from reactive repair to Condition-Based Maintenance (CBM). Below are 6 primary industrial domains where this platform can be directly deployed:

- **Automotive Stamping & Press Lines:** Heavy mechanical-hydraulic stamping presses exert 500+ tons per stroke. Detecting cylinder displacement lag and valve power spikes early prevents costly die damage, stamping defects, and sudden press jamming.
- **Industrial Injection Molding & Metal Extrusion:** Continuous metal extrusion machines require synchronized bilateral hydraulic force. Monitoring power and displacement balance between left and right modules flags piston seal wear and fluid thermal breakdown before seal blowouts occur.
- **Aerospace Flight Control Rigs & Ground Test Benches:** Flight control hydraulic actuators require zero-failure reliability. The Siamese network's similarity metric flags subtle mechanical anomalies in test rigs weeks before physical failure occurs.
- **Oil, Gas & Offshore Subsea Hydraulic Units (HPUs):** Hydraulic units operating subsea Blowout Preventers (BOPs) operate in harsh, remote environments where unplanned downtime costs over $1M/day. Early warning of solenoid valve drop allows planned offshore servicing.
- **Heavy Construction & Mining Mobile Machinery:** Excavators and tunnel boring machines face extreme shock and dirt contamination. Real-time telemetry monitoring prevents catastrophic main hydraulic pump cavitation and hydraulic cylinder scoring.
- **Renewable Energy (Hydroelectric & Wind Turbine Governors):** Hydraulic governors modulate water flow in hydro turbines, and pitch cylinders angle wind turbine blades. Automated fault monitoring prevents dangerous turbine runaway events and maximizes continuous energy generation.

5. Future Enhancements & Technological Roadmap

The current codebase establishes an end-to-end operational pipeline. To scale into enterprise industrial deployments, the following technological enhancements represent the strategic product roadmap:

- **Multi-Class & Hierarchical Fault Diagnosis:** Extend binary classification (0 vs 1) into granular multi-class fault categorization: classifying specific sub-assembly failure modes (e.g., valve stick-slip, internal fluid bypass, pump cavitation, cylinder seal scoring, accumulator pressure discharge).
- **Remaining Useful Life (RUL) Prognostics:** Incorporate temporal regression heads (Temporal Convolutional Networks or Survival Analysis) that output the estimated operating hours or cycles remaining until functional failure, empowering automated replacement scheduling.
- **Edge AI Model Quantization (INT8 / TensorRT / ONNX):** Quantize models to INT8 precision and export to ONNX / TensorRT for execution on embedded industrial edge hardware (NVIDIA Jetson, Siemens IPCs, Raspberry Pi 5) with sub-millisecond local inference and no cloud dependency.
- **Multi-Modal Sensor Fusion (Acoustic & Vibration Telemetry):** Integrate vibration accelerometers and high-frequency acoustic emission signals (as researched in notebooks/PM-CRNN.ipynb) using cross-attention transformers to detect micro-cracks before electrical/hydraulic deviations emerge.
- **Self-Supervised Industrial Foundation Models:** Pre-train masked autoencoder time-series transformers (PatchTST, TimesNet, or Chronos) on massive unlabeled industrial fleet telemetry, enabling fine-tuning on new production lines with minimal labeled historical failure data.
- **Generative AI Maintenance Copilot & Automated Work Orders:** Feed SHAP feature attributions into a domain-specific LLM agent to automatically generate plain-language diagnostic tickets, root-cause explanations, and recommended replacement part orders for shop-floor technicians.
- **Industrial IoT Protocols & Closed-Loop SCADA Integration:** Implement native OPC-UA and MQTT connectors to interface directly with factory PLCs (Siemens S7, Allen-Bradley), allowing automated closed-loop safety throttling or graceful machine shutdown when fault risk crosses critical thresholds.

6. Catalog of Models & Algorithms Across Codebase

Below is the complete inventory of all neural network models, statistical algorithms, preprocessing routines, and deployment frameworks utilized across both the core production system (`src/`) and exploratory research notebooks (`notebooks/`):

Model / Algorithm	Category	File Location	Role & Functionality in Project
1D-CNN (Convolutional Neural Net)	Deep Learning	src/cnn_model.py	Primary supervised classifier with 3 Conv1d blocks + 3 dense layers; detects temporal fault signatures.
Siamese Neural Network	Metric Learning	src/siamese_model.py	Twin Conv1D encoders mapping telemetry into 128-d metric space; evaluates Euclidean distance for anomaly similarity.
CRNN (Conv1D + Bidirectional GRU)	Hybrid DL / Research	notebooks/PM-CRNN.ipynb	Explores spatio-temporal modeling combining 1D convolutions with recurrent gating (GRU) and audio/frequency spectrograms.
1D-ResNet (Residual CNN)	Deep Learning / Research	notebooks/PMResnet.ipynb	Experiments with deep residual skip connections (residual_block) to prevent gradient degradation on long sequences.
SHAP (KernelExplainer)	Explainable AI (XAI)	src/feature_analysis.py	Game-theoretic feature attribution calculating SHAP values across time steps to rank individual sensor importance.
t-SNE (t-Distributed Stochastic Neighbor)	Dimensionality Reduction	src/visualization.py src/train.py	Projects high-dimensional latent layer activations into 2D plots to visualize separation between healthy and faulty states.
StandardScaler & Sequence Windowing	Data Preprocessing	src/data_processor.py	Applies Z-score scaling and sliding window segmentation to convert continuous time logs into (N, 8, 18) sequences.
Pairwise Siamese Generator	Dataset Algorithm	src/data_processor.py	Generates balanced positive pairs (label 1: same state) and negative pairs (label 0: contrasting state) for contrastive training.
ROC-AUC & Confusion Matrix	Statistical Evaluation	src/evaluation.py src/cnn_model.py	Calculates classification accuracy, precision, recall, F1-scores, decision threshold curves, and confusion heatmaps.
FastAPI Async Inference Server	Backend API	src/api/app.py	Provides high-performance RESTful microservice with lifespan model loading, health checks, and JSON /predict endpoints.
Streamlit Reactive Web App	User Interface	app.py	Single-file interactive UI featuring single-sample manual input, sliding-window CSV upload, and model evaluation plot gallery.

Architectural Takeaway: The repository demonstrates a complete engineering evolution: from exploratory research in Jupyter notebooks (evaluating CRNNs, ResNets, and spectrograms) to a lean, production-grade PyTorch implementation. The core 1D-CNN delivers rapid, deterministic fault classification, while the Siamese Network provides robust baseline-comparison anomaly scoring, backed by full SHAP explainability, FastAPI microservices, and an interactive Streamlit UI.